$$= \frac{3}{11} \text{ of } \frac{8x}{11} = \frac{24x}{121}.$$

$\therefore \quad \frac{3x}{11} - \frac{24x}{121} = 9 \Leftrightarrow 33x - 24x = 9 \times 121$

$\Leftrightarrow 9x = 9 \times 121 \Leftrightarrow x = 121.$

292. Let the hourly wage of each sales person be ₹ x.
Then, hourly wage of each mechanic = ₹ $(2x)$.

Hourly wage of each custodial worker = ₹ $\left(\frac{2x}{3}\right)$.

$\therefore$ Required fraction = $\frac{x}{\left(\frac{2x}{3}\right)} = \frac{3}{2}$.

293. Sukhvinder's monthly income = ₹ $\left(\frac{234000}{12}\right)$ = ₹ 19500.

Jassi's monthly income = ₹ $\left(\frac{3}{2} \times 19500\right)$ = ₹ 29250.

$\therefore$ Ganeshi's monthly income = ₹ (2×29250) = ₹ 58500.

294. Cost per kg of rice last year = ₹ $\left(\frac{1044}{36}\right)$ = ₹ 29.

Cost per kg of rice this year = ₹ $\left(\frac{768}{24}\right)$ = ₹ 32.

$\therefore$ Required difference = ₹ $(32 - 29)$ = ₹ 3.

295. $12B + 30W = 8940 \Rightarrow \frac{1}{3}(12B + 30W) = \frac{1}{3} \times 8940$

$\Rightarrow 4B + 10W = 2980.$

296. $5P + 8C = 145785 \Rightarrow 3(5P + 8C) = 3 \times 145785$

$\Rightarrow 15P + 24C = 437355.$

297. $21T + 35C = 41825 \Rightarrow \frac{3}{7}(21T + 35C) = \frac{3}{7} \times 41825$

$\Rightarrow 9T + 15C = 17925.$

298. Cost of 1 kg of sugar = ₹ $\left(\frac{195}{13}\right)$ = ₹ 15.

Cost of 1 kg of rice = ₹ $\left(\frac{544}{17}\right)$ = ₹ 32.

Cost of 1 kg of wheat = ₹ $\left(\frac{336}{21}\right)$ = ₹ 16.

$\therefore$ Required cost = ₹ $(15 \times 21 + 32 \times 26 + 16 \times 19)$
= ₹ $(315 + 832 + 304)$ = ₹ 1451.

299. (*a*) Price per kg = ₹ $\left(\frac{160}{10}\right)$ = ₹ 16;

(*b*) Price per kg = ₹ $\left(\frac{30}{2}\right)$ = ₹ 15;

(*c*) Price per kg = ₹ $\left(\frac{70}{4}\right)$ = ₹ 17.50;

(*d*) Price per kg = ₹ $\left(\frac{340}{20}\right)$ = ₹ 17;

(*e*) Price per kg = ₹ $\left(\frac{130}{8}\right)$ = ₹ 16.25.

Clearly, the most economical price is 2 kilo for ₹ 30.

300. Let the time spent on each Maths problem be $(2x)$ minutes and that spent on each other question be x minutes.
Then, $50 \times 2x + 150x = 3 \times 60 \Leftrightarrow 250x = 180$

$\Leftrightarrow x = \frac{180}{250} = \frac{18}{25}.$

$\therefore$ Total time spent on Mathematics problems

$= \left(50 \times 2 \times \frac{18}{25}\right) \text{min} = 72 \text{ min}.$

301. Duration from 4.56 P.M. to 5.32 P.M. = 36 min.

$= \left(\frac{36}{60}\right) \text{hour} = \frac{3}{5} \text{hour}.$

302. 72 hr 6 min = $(72 \times 60 + 6)$ min = 4326 min.

$\therefore$ 72 hr 6 min ÷ 14 = 4326 min ÷ 14
= 309 min
= 5 hr 9 min.

$$\begin{array}{r} 309 \\ 14\overline{)4326} \\ \underline{42} \\ 126 \\ \underline{126} \\ \times \end{array}$$

303. Time between 10 a.m. and 13.27 hours = 3 hrs. 27 min. = 207 min.
Total duration of free time = (5×3) min = 15 min.
Remaining time = $(207 - 15)$ min. = 192 min.
$\therefore$ Duration of each of the 4 periods

$= \left(\frac{192}{4}\right) \text{min.} = 48 \text{ min}.$

304.

	Hrs.	Min.	Sec.
	3	17	49
(–)	1	54	50
	1	22	59

Total time = $(1 \times 60 + 22)$ min. + 59 sec.
= $(82 \times 60 + 59)$ sec.
= 4979 sec.

$\therefore$ Number of times the light is seen = $\left(\frac{4979}{13} + 1\right) = 384.$

305. The woman works from Sunday to Saturday and rests for the first time on Sunday. She then works from Monday to Sunday and rests for the second time on Monday. Likewise, she rests for the third time on Tuesday. Thus, starting from Sunday, for the 10th time she would rest on Tuesday. (as the cycle repeats after every 7 days)

306. $9 * 11 = 9^2 + (11)^2 - 9 \times 11 = 81 + 121 - 99 = 103.$

307. $(3 * -1) = \frac{3 \times (-1)}{3 + (-1)} = \frac{-3}{2}$. So, $3 * (3 * -1)$

$$= 3 * \left(\frac{-3}{2}\right) = \frac{3 \times \left(\frac{-3}{2}\right)}{3 + \left(\frac{-3}{2}\right)} = \frac{-9}{2} \times \frac{2}{3} = -3.$$

308. $3 * 5 + 5 * 3 = (2 \times 3 - 3 \times 5 + 3 \times 5) + (2 \times 5 - 3 \times 3 + 5 \times 3)$
$= (6 + 10 - 9 + 15) = 22.$

309. $4 \oplus (3 \oplus p) = 4 \oplus (3^2 + 2p) = 4 \oplus (9 + 2p) = 4^2 + 2(9 + 2p) = 34 + 4p.$

$\therefore \quad 34 + 4p = 50 \Rightarrow 4p = 50 - 34 = 16 \Rightarrow p = 4.$

310. $(a * b * c) * a * b = \left(\frac{a+b}{c}\right) * a * b = \frac{\left(\frac{a+b}{c}\right) + a}{b} = \frac{a + b + ac}{bc}.$